AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 9 · WORKBOOK

Features and Distractors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected feature-selection record.

Define inputs, construct collisions, expose a colour shortcut and reconsider the feature set when the target changes.

Keep one habit: Keep the task, units, available inputs, expected labels and observation status together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Operational Labels

Debugging Strategies

Branching Logic

W1 · Separate permitted inputs from the target

> Write a defined feature order and units.

Your evidence record

- Card reference for W1–W3: positive whole-number length and width in centimetres; fixed flat orientation. Fits means length <=12 AND width <=8, including equality; otherwise too large. Choose inputs available before inspection. The final inspection sticker is added afterward. For W5 the target changes to red/blue bins, so review feature relevance again.

- For the fixed-orientation card task, fits requires length at most 12 cm and width at most 8 cm, including equality.

  Choose the measurements needed as **input**s from length, width, colour, card number and the final inspection sticker. State the **target label** and write the **feature record** for a 12 cm by 8 cm red card using a defined order.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Demonstrate a width-only collision

> Use equal widths and lengths on opposite sides of the limit.

Your evidence record

- Card reference for W1–W3: positive whole-number length and width in centimetres; fixed flat orientation. Fits means length <=12 AND width <=8, including equality; otherwise too large. Choose inputs available before inspection. The final inspection sticker is added afterward. For W5 the target changes to red/blue bins, so review feature relevance again.

- Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels. Explain why a system receiving only width cannot always label both correctly.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Break the colour shortcut

> Use measurements to justify the correct label.

Your evidence record

- Card reference for W1–W3: positive whole-number length and width in centimetres; fixed flat orientation. Fits means length <=12 AND width <=8, including equality; otherwise too large. Choose inputs available before inspection. The final inspection sticker is added afterward. For W5 the target changes to red/blue bins, so review feature relevance again.

- A practice set contains three blue cards that fit and three red cards that are too large. A helper uses blue means fits and red means too large. Create a new card that this helper mislabels.

  Explain why the perfect practice score did not establish that colour was a reliable **feature** for this task.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Reconsider inputs when the target changes

> The same dimensions can now require different colour-bin labels.

Your evidence record

- Card reference for W1–W3: positive whole-number length and width in centimetres; fixed flat orientation. Fits means length <=12 AND width <=8, including equality; otherwise too large. Choose inputs available before inspection. The final inspection sticker is added afterward. For W5 the target changes to red/blue bins, so review feature relevance again.

- The delivery station now sorts cards into red and blue bins instead of checking dimensions. Which previously unneeded **feature** becomes useful? Explain why length and width alone may now be insufficient.

  Give two cards that support your explanation.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Changed independent case

> Keep one first attempt before hints or feedback.

Read and explain independently

Training covers show space books only on yellow backgrounds and animal books only on blue backgrounds. Design two controlled comparisons to test whether a classifier uses the intended subject.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Use explainable traits that connect to the reader's stated preferences.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Prove that one measurement is insufficient for **Feature**s and useful distinctions.
Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

All animal books in **training** have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

▼ Extra practice · A complete case

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful **evidence** and possible distractions.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Predict whether a book is short using page count and cover colour. Which **feature** directly supports the defined label? Give a counterexample to colour.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 9 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

All animal books in **training** have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Extra practice · Display-tray features

REFERENCE · SECTION 15

Extra practice · Display-tray features

> This is a separate optional transfer task rather than a duplicate of selected W4.

The new rule, counterexample and timing question

- A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

  A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

  Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

Training contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

AI ACADEMY / AI-2/9

Features and Distractors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Spot confounding

Topic and logo always move together in the training set.

2. Hold topic fixed

Compare the same mystery description with triangle versus circle logo.

3. Hold logo fixed

Compare mystery versus science descriptions with identical triangle logo.

4. Observe separately

Record predictions before concluding which clue the classifier uses.

CHANGE ONE THING

The first controlled pair yields mystery for triangle and science for circle.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.
- Which labelled step matches this detail? Record **prediction**s before concluding which clue the classifier uses.
- 1. Hold topic fixed
- 2. Hold logo fixed
- 3. Observe separately
- 4. Spot confounding

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Observe separately Record **prediction**s before concluding which clue the classifier uses.

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Changed-case question

- Changed-case practice: The first controlled pair yields mystery for triangle and science for circle. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. That pair is **evidence** of unwanted logo sensitivity.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That pair is **evidence** of unwanted logo sensitivity. The subject description stayed mystery; only the ir**relevant** logo changed.

Repair and save your work

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality boundary for Short..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

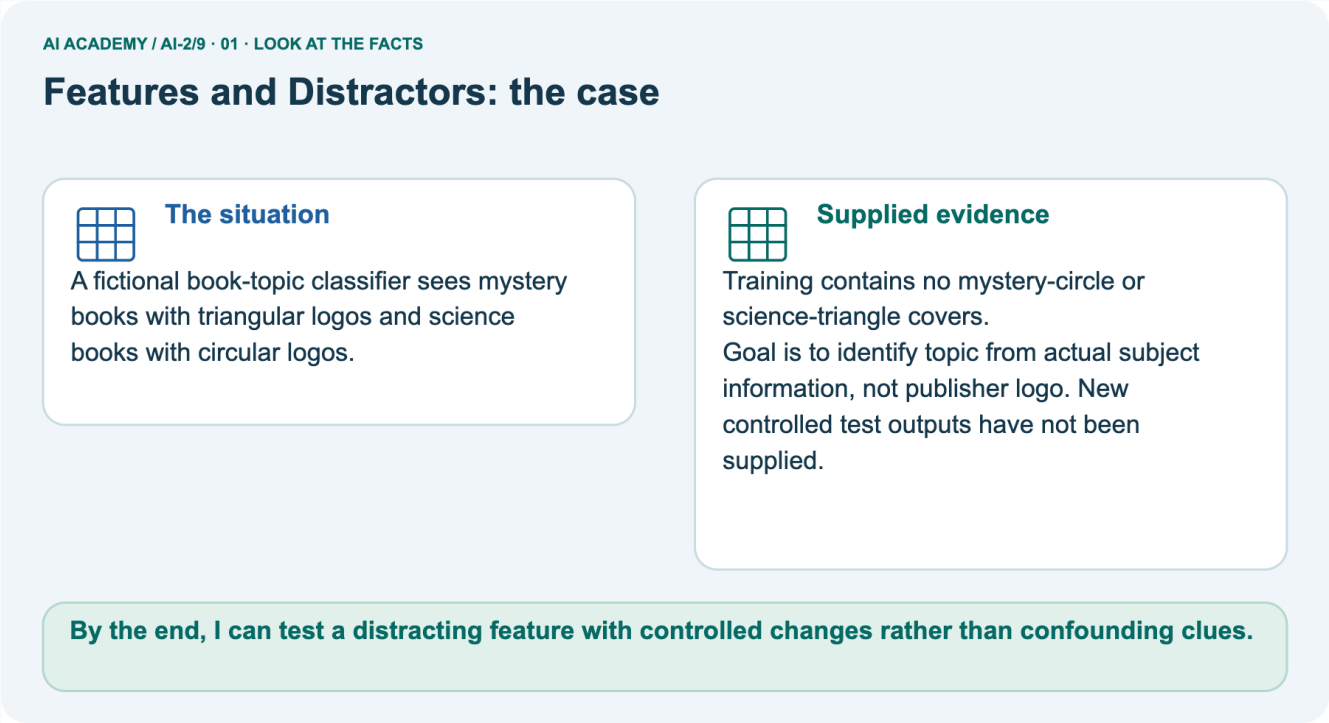
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Features and Distractors: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

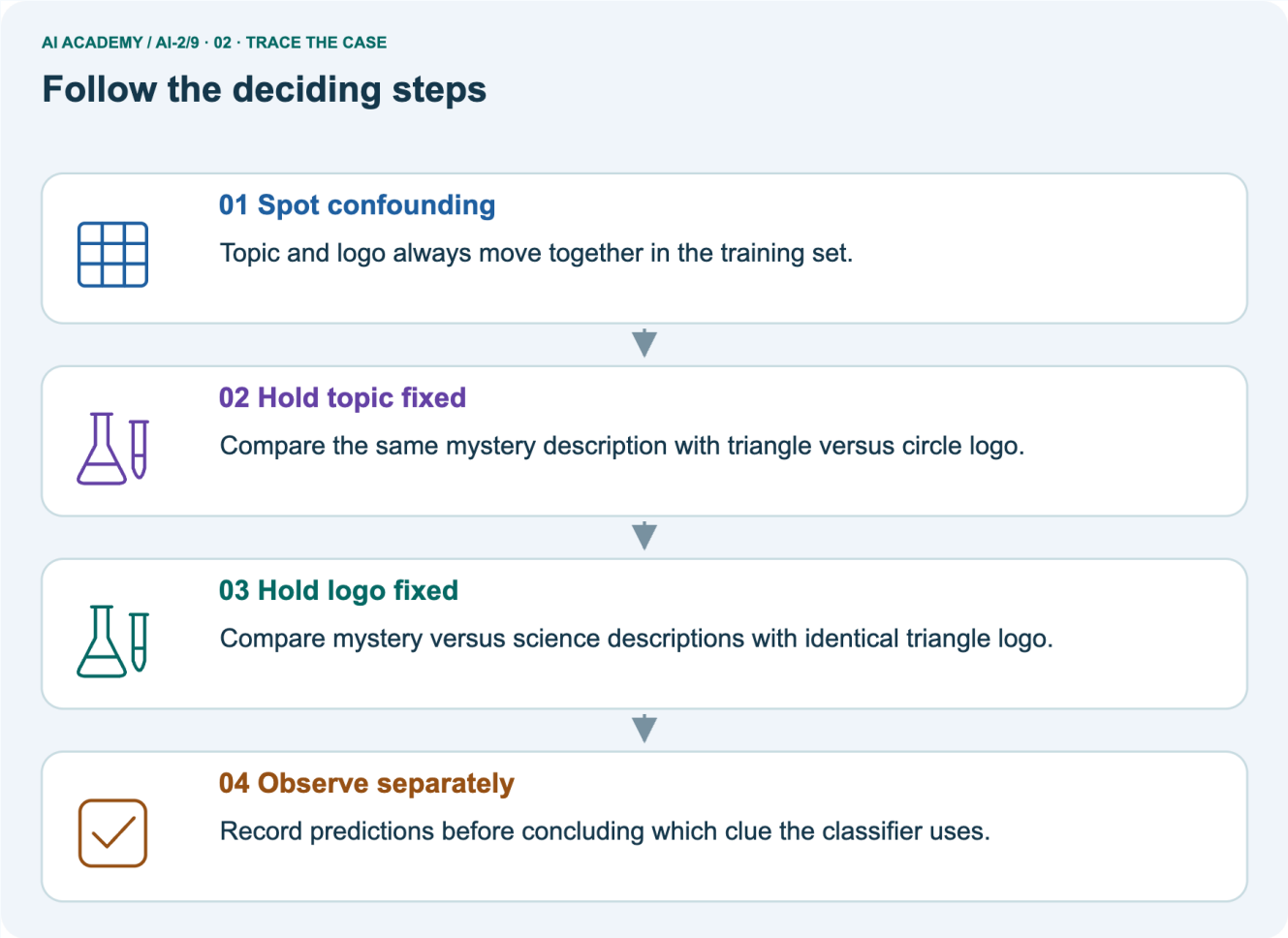
A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos. Supplied facts: **Training** contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



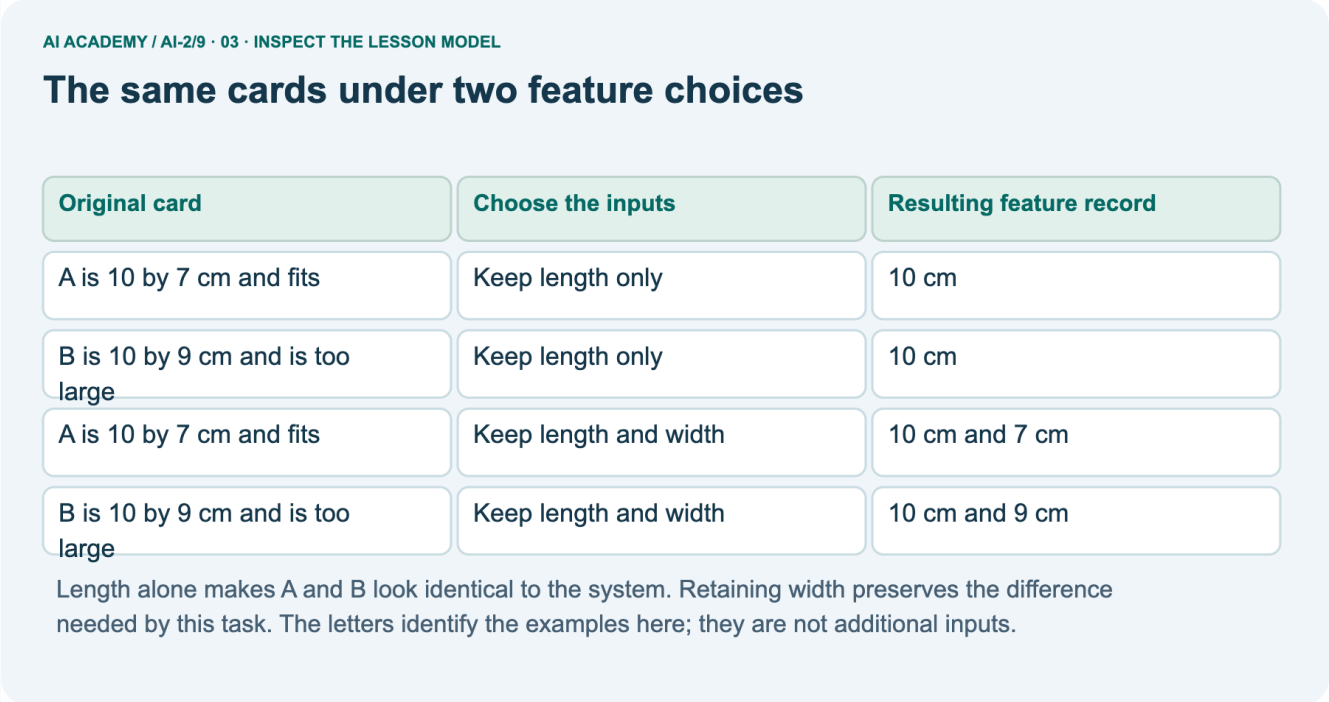
Follow the labelled picture with the matching explanation. Open the vector diagram

Spot confounding: Topic and logo always move together in the **training** set. → Hold topic fixed: Compare the same mystery description with triangle versus circle logo. → Hold logo fixed: Compare mystery versus science descriptions with identical triangle logo. → Observe separately: Record **prediction**s before concluding which clue the classifier uses.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. The same cards under two feature choices



AI ACADEMY / AI-2/9 · 03 · INSPECT THE LESSON MODEL

The same cards under two feature choices

Original card	Choose the inputs	Resulting feature record
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

Follow the labelled picture with the matching explanation. Open the vector diagram

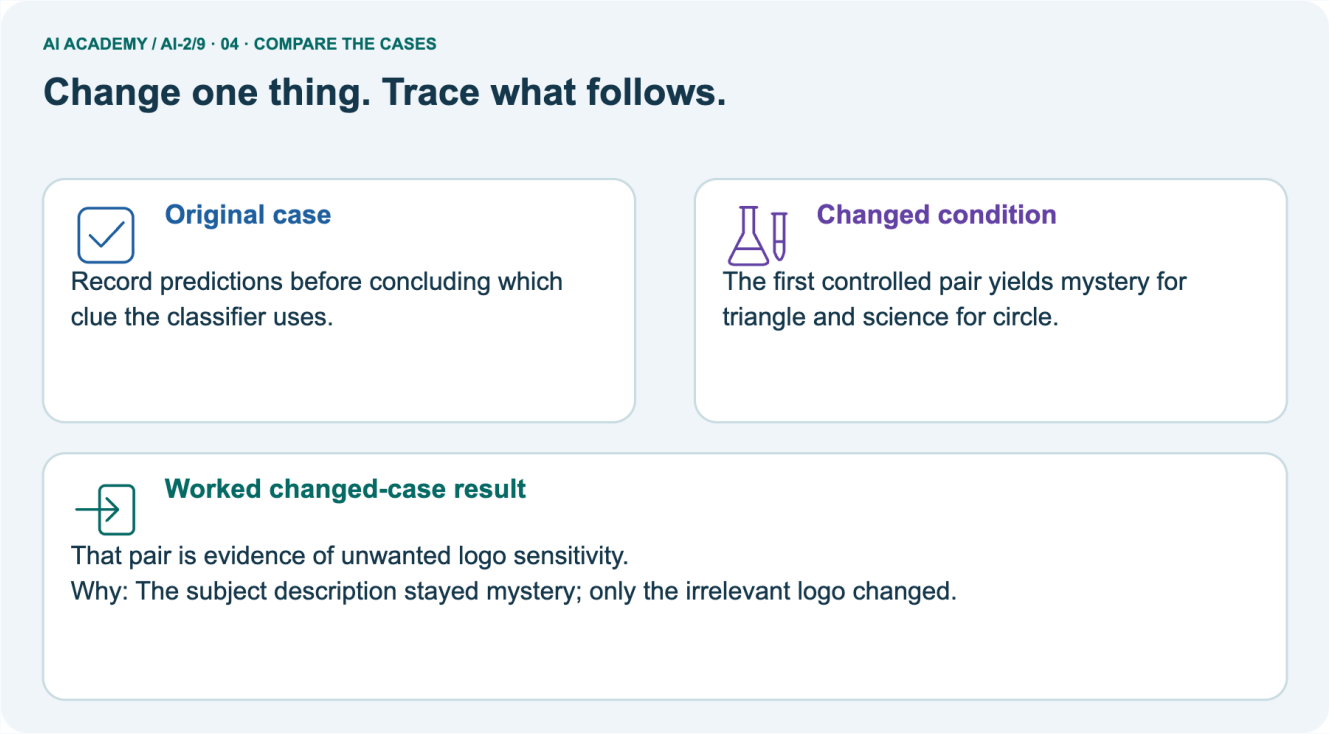
The same cards under two **feature** choices. A is 10 by 7 cm and fits | Keep length only | 10 cm. B is 10 by 9 cm and is too large | Keep length only | 10 cm. A is 10 by 7 cm and fits | Keep length and width | 10 cm and 7 cm.

B is 10 by 9 cm and is too large | Keep length and width | 10 cm and 9 cm

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

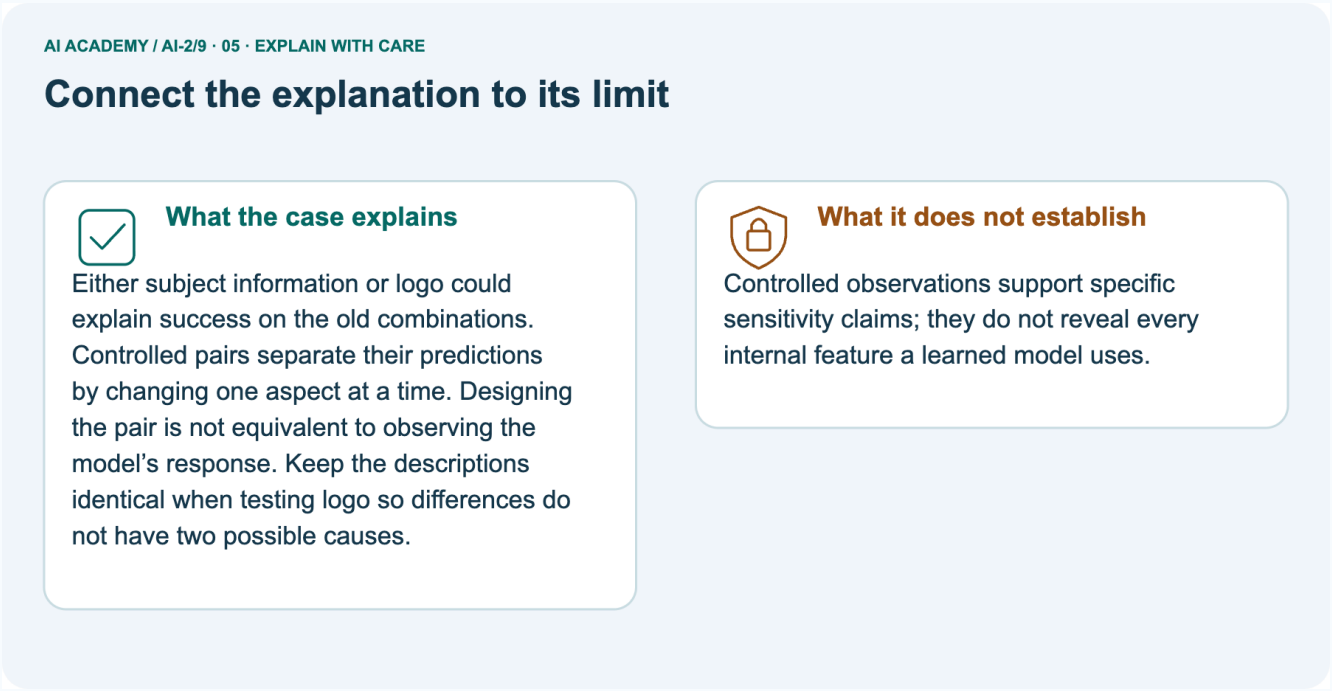
Original case: Record **prediction**s before concluding which clue the classifier uses. Changed condition: The first controlled pair yields mystery for triangle and science for circle.

Expected result: That pair is **evidence** of unwanted logo sensitivity. Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

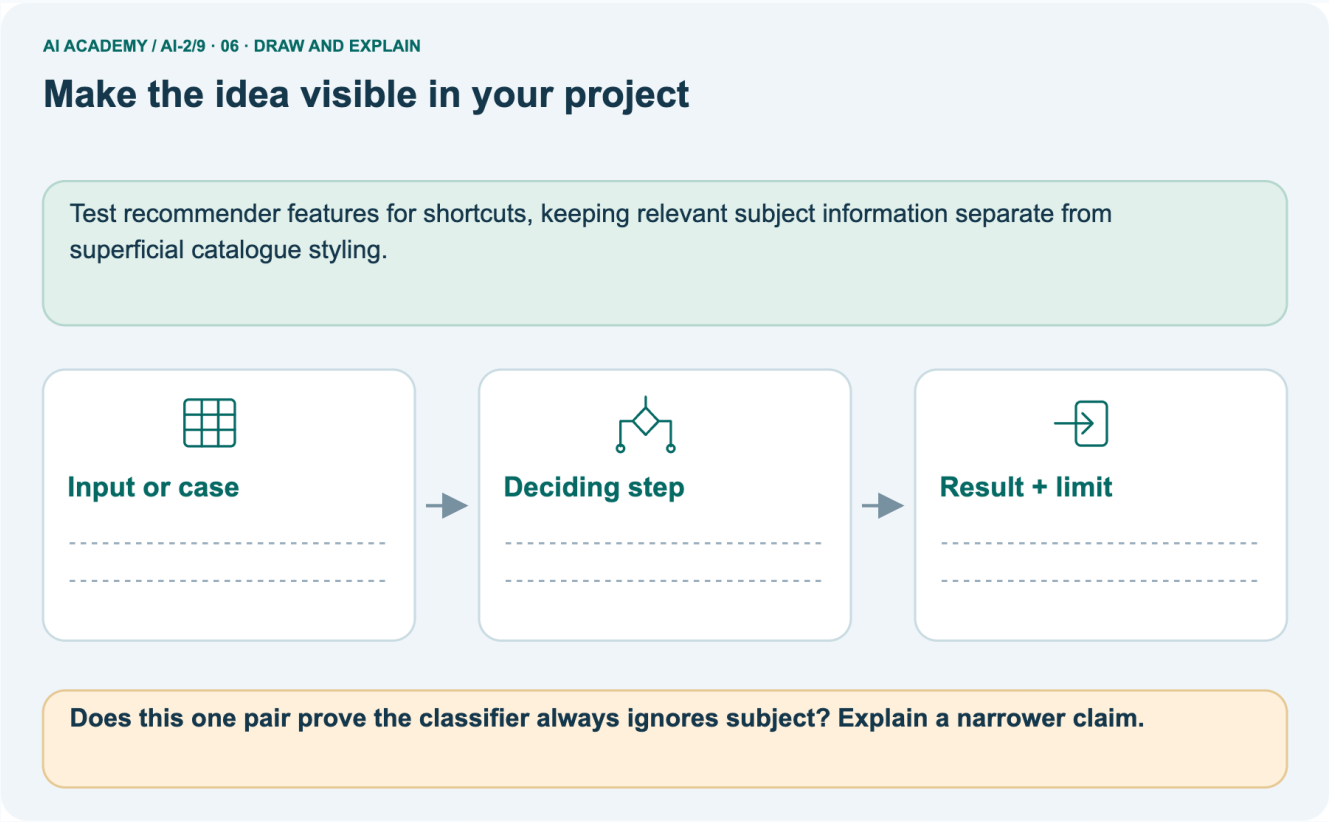
Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Limit: Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling. Draw your **input**, deciding step, result and limit.

Supported question: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Features and Distractors: the case

02. Follow the deciding steps

03. The same cards under two feature choices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.